LAB 10

Cards, Lists, Grids, and Slivers

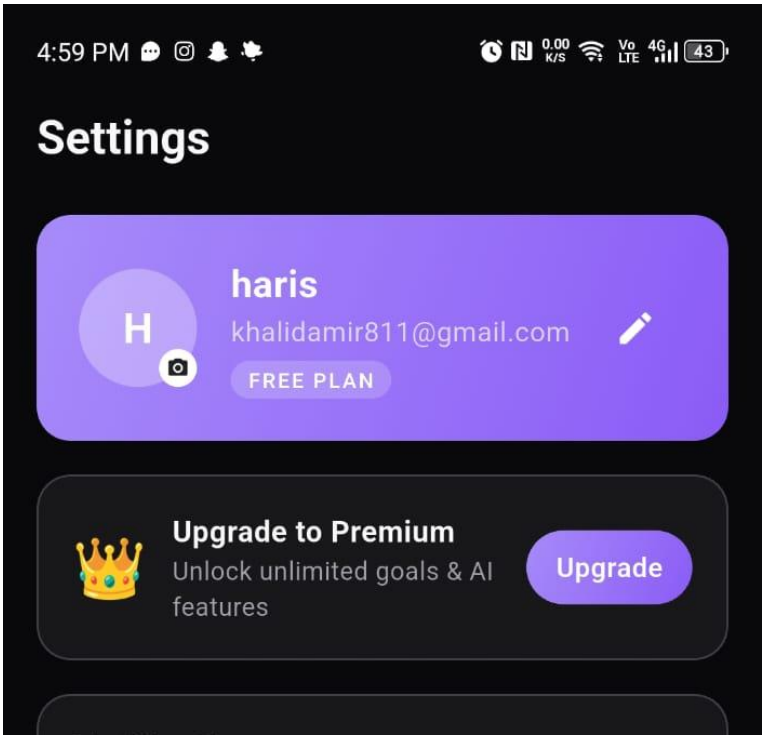
1. CARDS:

Setting screen

```
SliverToBoxAdapter(
  child: Padding(
    padding: const EdgeInsets.fromLTRB(16, 8, 16, 16),
    child: Container(
      decoration: BoxDecoration(
        gradient: theme.linearGradient,
        borderRadius: BorderRadius.circular(16),
      ),
      padding: const EdgeInsets.all(20),
      child: Row(
        children: [
          GestureDetector(
            onTap: _showUploadDialog,
            onLongPress: _showUploadDialog,
            child: Stack(
              children: [
                CircleAvatar(
                  radius: 28,
                  backgroundColor: Colors.white.withValues(alpha: 0.3),
                  backgroundImage: _localImage != null
                      ? FileImage(_localImage!) as ImageProvider
                      : (photoUrl != null ? NetworkImage(photoUrl) : null),
                  child: (_localImage == null && photoUrl == null)
                      ? Text(
                          initials,
                          style: const TextStyle(
                            color: Colors.white,
                            fontSize: 20,
                            fontWeight: FontWeight.bold,
                          ),
```

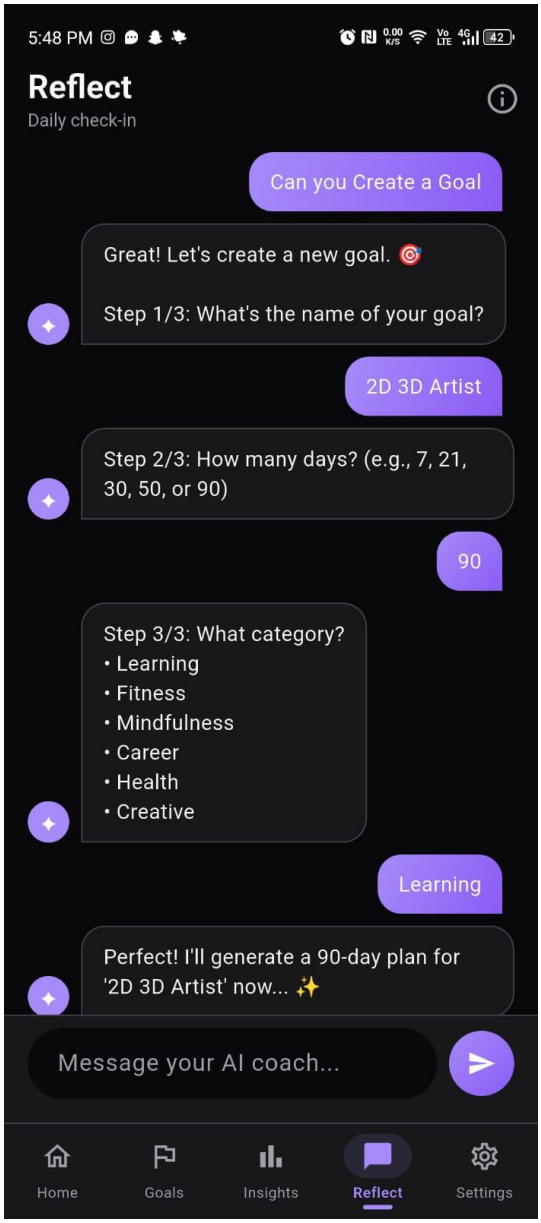
```
                ),
                Positioned(
                  bottom: 0,
                  right: 0,
                  child: Container(
                    width: 18,
                    height: 18,
                    decoration: const BoxDecoration(
                      color: Colors.white,
                      shape: BoxShape.circle,
                    ),
                    child: const Icon(Icons.camera_alt, size: 11, color: Colors.black87),
                  ),
                ),
              ],
            ),
          ),
          const SizedBox(width: 16),
          Expanded(
            child: Column(
              crossAxisAlignment: CrossAxisAlignment.start,
              children: [
                Text(
                  name,
                  style: const TextStyle(
                    color: Colors.white,
                    fontSize: 18,
                    fontWeight: FontWeight.bold,
                  ),
                ),
                Text(
                  email,
                  style: const TextStyle(
```

```
                    style: const TextStyle(
                        color: Colors.white70, fontSize: 13),
                  ),
                  const SizedBox(height: 4),
                  Container(
                    padding: const EdgeInsets.symmetric(
                        horizontal: 8, vertical: 2),
                    decoration: BoxDecoration(
                      color: Colors.white.withValues(alpha: 0.2),
                      borderRadius: BorderRadius.circular(10),
                    ),
                    child: const Text(
                      'FREE PLAN',
                      style: TextStyle(
                          color: Colors.white,
                          fontSize: 10,
                          fontWeight: FontWeight.w600,
                          letterSpacing: 1),
                    ),
                  ),
                ],
              ),
            ),
            IconButton(
              icon: const Icon(Icons.edit, color: Colors.white,
              size: 20),
              onPressed: _showUploadDialog,
            ),
          ],
        ),
      ),
    ),
  ),
```



Reflect screen:

```
Container(
  padding: const EdgeInsets.symmetric(
      horizontal: 14, vertical: 10),
  decoration: BoxDecoration(
    gradient: isUser
        ? theme.linearGradient
        : null,
    color: isUser ? null : theme.card,
    borderRadius: BorderRadius.only(
      topLeft: const Radius.circular(16),
      topRight: const Radius.circular(16),
      bottomLeft: isUser
          ? const Radius.circular(16)
          : Radius.zero,
      bottomRight: isUser
          ? Radius.zero
          : const Radius.circular(16),
    ),
    border: isUser
        ? null
        : Border.all(color: theme.border),
  ),
  child: Text(
    text,
    style: TextStyle(
      color: isUser
          ? Colors.white
          : theme.text,
      fontSize: 14,
    ),
  ),
),
```



Cards:

- Gradient profile card at top of Settings -> photo, name, email, FREE PLAN badge.
- Chat bubbles: user messages have gradient; AI messages have card color.

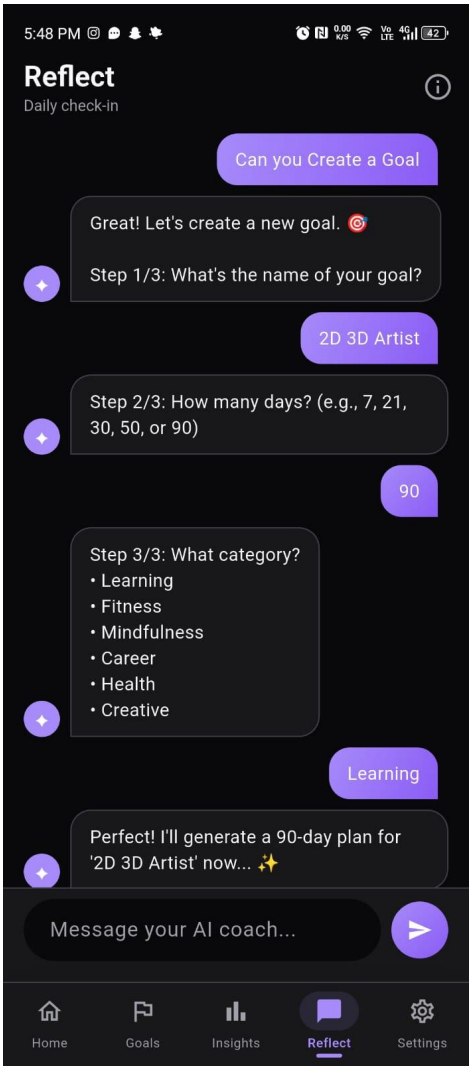
2. LISTS:

```
304          Expanded(
305            child: ListView.builder(
306              controller: _scrollCtrl,
307              padding: const EdgeInsets.symmetric(horizontal: 16, vertical: 8),
308              itemCount: _messages.length + (_typing ? 1 : 0),
309              itemBuilder: (context, index) {
310                if (_typing && index == _messages.length) {
311                  return _TypingIndicator(theme: theme);
312                }
313                final msg = _messages[index];
314                final isUser = msg['isUser'] as bool? ?? false;
315                final text = msg['text'] as String? ?? '';
316                final id = msg['id'] as String? ?? '';
317                final selected = _selected.contains(id);
318
319                return GestureDetector(
320                  onLongPress: () => _toggleSelect(id),
321                  onTap: _selectMode ? () => _toggleSelect(id) : null,
322                  child: AnimatedContainer(
323                    duration: const Duration(milliseconds: 200),
324                    color: selected
325                        ? theme.accent.withValues(alpha: 0.1)
326                        : Colors.transparent,
327                    child: Padding(
328                      padding: const EdgeInsets.only(bottom: 8),
329                      child: Row(
330                        mainAxisAlignment: isUser
331                            ? MainAxisAlignment.end
332                            : MainAxisAlignment.start,
333                        crossAxisAlignment: CrossAxisAlignment.end,
334                        children: [
335                          if (!isUser) ...[
336                            CircleAvatar(
```

```
336                            CircleAvatar(
337                              radius: 14,
338                              backgroundColor: theme.accent,
339                              child: const Text('✦',
340                                  style: TextStyle(
341                                      color: Colors.white, fontSize: 12)),
342                            ),
343                            const SizedBox(width: 8),
344                          ],
345                          Flexible(
346                            child: Column(
347                              crossAxisAlignment: isUser
348                                  ? CrossAxisAlignment.end
349                                  : CrossAxisAlignment.start,
350                              children: [
351                                if (!isUser &&
352                                    (msg['showLabel'] as bool? ?? false))
353                                  Padding(
354                                    padding: const EdgeInsets.only(bottom: 4),
355                                    child: Text(
356                                      'PERSIST AI',
357                                      style: TextStyle(
358                                          color: theme.accent,
359                                          fontSize: 10,
360                                          fontWeight: FontWeight.bold,
361                                          letterSpacing: 1),
362                                    ),
363                                  ),
364                                Container(
365                                  padding: const EdgeInsets.symmetric(
366                                      horizontal: 14, vertical: 10),
367                                  decoration: BoxDecoration(
368                                    gradient: isUser
369                                        ? theme.linearGradient
```

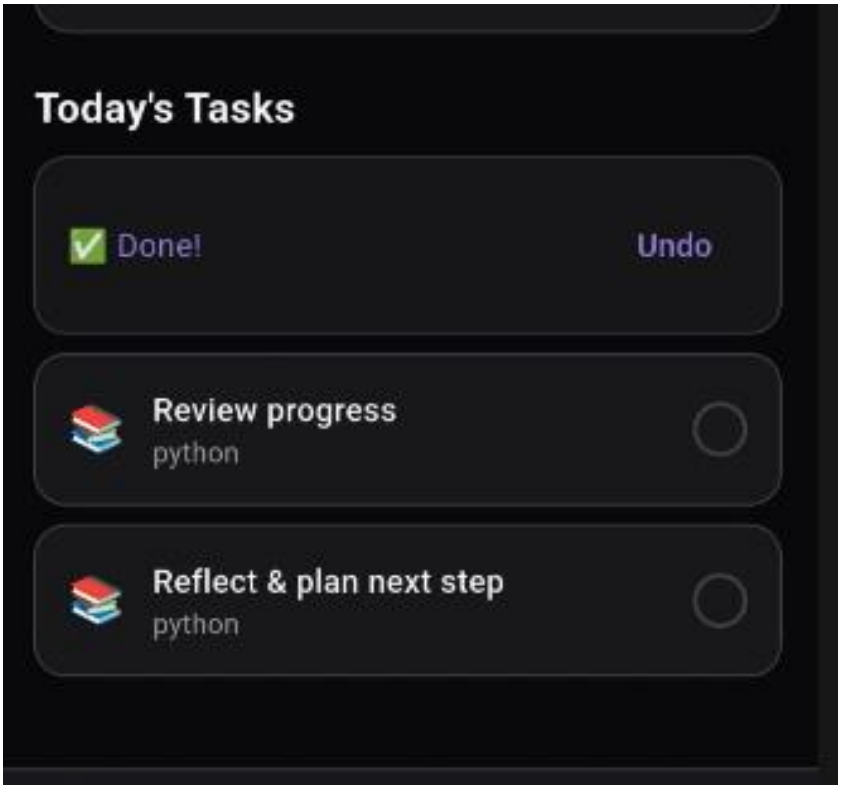
```
                    : null,
                  color: isUser ? null : theme.card,
                  borderRadius: BorderRadius.only(
                    topLeft: const Radius.circular(16),
                    topRight: const Radius.circular(16),
                    bottomLeft: isUser
                        ? const Radius.circular(16)
                        : Radius.zero,
                    bottomRight: isUser
                        ? Radius.zero
                        : const Radius.circular(16),
                  ),
                  border: isUser
                      ? null
                      : Border.all(color: theme.border),
                ),
                child: Text(
                  text,
                  style: TextStyle(
                    color: isUser
                        ? Colors.white
                        : theme.text,
                    fontSize: 14,
                  ),
                ),
              ),
              if (selected)
                Padding(
                  padding: const EdgeInsets.only(top: 4),
                  child: Icon(Icons.check_circle,
                      size: 16, color: theme.accent),
                ),
            ],
          ),
```

```
                  ),
                ],
              ),
            ),
            if (isUser) const SizedBox(width: 8),
          ],
        ),
      ),
    ),
  );
},
),
),
```


5:48 PM
Reflect
Daily check-in
Can you Create a Goal
Great! Let's create a new goal. 🎯
Step 1/3: What's the name of your goal?
2D 3D Artist
Step 2/3: How many days? (e.g., 7, 21, 30, 50, or 90)
90
Step 3/3: What category?
• Learning
• Fitness
• Mindfulness
• Career
• Health
• Creative
Learning
Perfect! I'll generate a 90-day plan for '2D 3D Artist' now... ✨
Message your AI coach...
Home
Goals
Insights
Reflect
Settings

```
334                 SliverList(
335                   delegate: SliverChildBuilderDelegate(
336                     (context, index) {
337                       final entry = tasks[index];
338                       final task = entry['task'];
339                       final isUndo = _pendingUndoTaskId == task.id;
340
341                       return Padding(
342                         padding: const EdgeInsets.fromLTRB(16, 0, 16, 8),
343                         child: AnimatedOpacity(
344                           opacity: task.done && !isUndo ? 0.5 : 1.0,
345                           duration: const Duration(milliseconds: 300),
346                           child: Container(
347                             padding: const EdgeInsets.all(14),
348 >                           decoration: BoxDecoration(···
352                             ),
353                             child: isUndo
354                                 ? Row(
355                                     children: [
356 >                                   Text('✅ Done!',···
358                                     const Spacer(),
359 >                                   TextButton(···
363                                     ),
364                                     ],
365                                   )
366                                 : Row(
367                                     children: [
368                                     Text(entry['categoryEmoji'] ?? '🎯',
369                                         style: const TextStyle(fontSize: 20)),
370                                     const SizedBox(width: 12),
371                                     Expanded(
372                                       child: Column(
373                                         crossAxisAlignment: CrossAxisAlignment.start,
```

```
374                                         children: [
375                                           Text(
376                                             task.text,
377 >                                         style: TextStyle(···
384                                           ),
385                                           ),
386                                           Text(
387                                             entry['goalName'] ?? '',
388 >                                         style: TextStyle(···
391                                           ),
392                                           ),
393                                         ],
394                                       ),
395                                     ),
396                                     GestureDetector(
397 >                                     onTap: task.done···
400 >                                     child: AnimatedContainer(···
420                                       ),
421                                     ),
422                                   ],
423                                 ),
424                           ),
425                         ),
426                       );
427                     },
428                     childCount: tasks.length,
429                   ),
430                 ),
431
432               const SliverToBoxAdapter(child: SizedBox(height: 32)),
```



Lists:

- Reflect screen with Scrollable chat list.
- Each item is a message bubble.
- Auto-scrolls to bottom on new message.
- Today's task list on Home screen
- Only renders visible items.

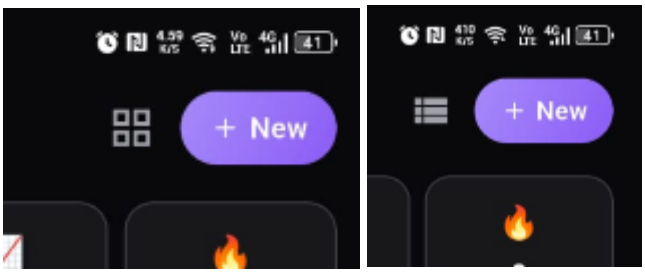
3. GRIDS

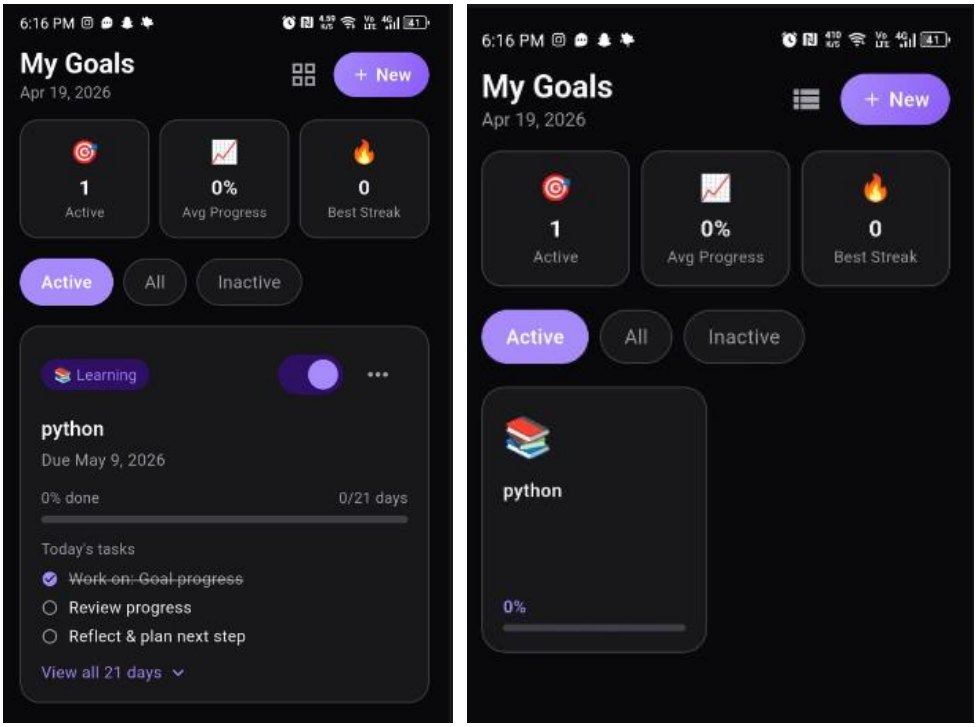
Goal Screen:

```
74                IconButton(
75                  icon: Icon(
76                    _isGrid ? Icons.view_list : Icons.grid_view,
77                    color: theme.textMuted,
78                  ),
79                  onPressed: () => setState(() => _isGrid = !_isGrid),
80                ),
```

```
186            SliverPadding(
187              padding: const EdgeInsets.symmetric(horizontal: 16),
188              sliver: SliverGrid(
189 >              gridDelegate: const
                   SliverGridDelegateWithFixedCrossAxisCount(…
194                ),
195                delegate: SliverChildBuilderDelegate(
196                  (context, index) {
197                    final goal = filtered[index];
198                    return GestureDetector(
199                      onTap: () => Navigator.push(context,
200                          MaterialPageRoute(builder: (_) => GoalDetailScreen
                           (goal: goal))),
201                      child: Container(
202                        padding: const EdgeInsets.all(14),
203 >                      decoration: BoxDecoration(…
207                        ),
208                        child: Column(
209                          crossAxisAlignment: CrossAxisAlignment.start,
210                          children: [
211                            Text(goal.categoryEmoji, style: const TextStyle
                             (fontSize: 28)),
212                            const SizedBox(height: 8),
213 >                          Text(…
218                            ),
219                            const Spacer(),
220                            TweenAnimationBuilder<double>(
221                              tween: Tween(begin: 0, end: goal.
                               progressPercent),
222                              duration: const Duration(milliseconds: 800),
223                              curve: Curves.easeOutCubic,
224                              builder: (_, value, __) => Column(
225                                crossAxisAlignment: CrossAxisAlignment.
                                 start,
```

```
226                                children: [
227 >                                Text('${(value * 100).toStringAsFixed(0)}
                                     %', …
229                                  const SizedBox(height: 4),
230                                  ClipRRect(
231                                    borderRadius: BorderRadius.circular(4),
232                                    child: LinearProgressIndicator(
233                                      value: value,
234                                      backgroundColor: theme.border,
235                                      valueColor: AlwaysStoppedAnimation
                                       (theme.accent),
236                                      minHeight: 5,
237                                    ),
238                                  ),
239                                ],
240                              ),
241                            ),
242                          ],
243                        ),
244                      ),
245                    );
246                  },
247                  childCount: filtered.length,
248                ),
249              ),
250            )
```





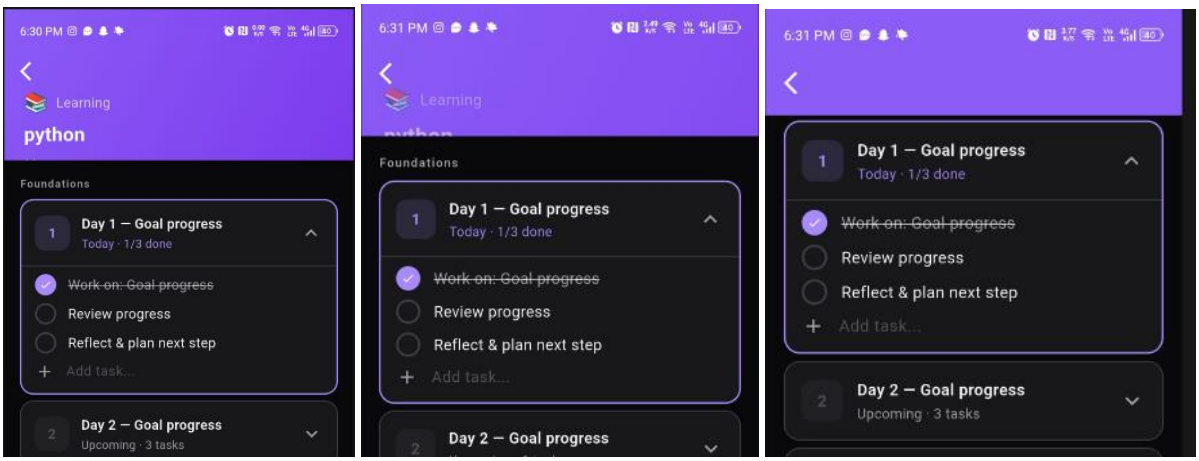
Grids

- Goals Tab tap the **grid icon** in top-right to switch from list to 2-column grid.
- Each cell shows emoji, goal name, animated progress bar.
- Tap list icon to go back.

4. SLIVERS:

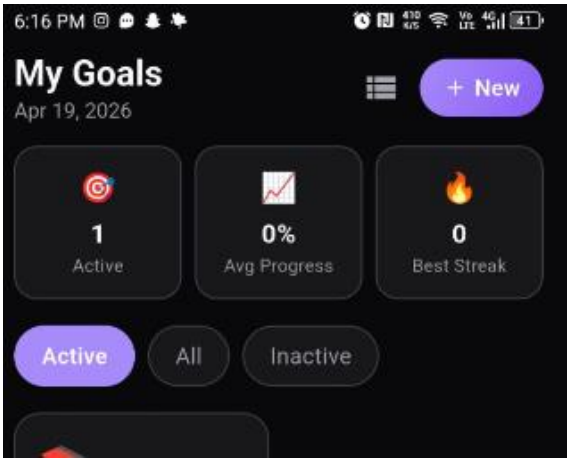
Goal details:

```
87          SliverAppBar(
88            expandedHeight: 160,
89            pinned: true,
90            flexibleSpace: FlexibleSpaceBar(
91              background: Container(
92                decoration: BoxDecoration(gradient: theme.headerGradient),
93                child: SafeArea(
94                  child: Padding(
95                    padding: const EdgeInsets.fromLTRB(20, 56, 20, 20),
96                    child: Column(
97                      crossAxisAlignment: CrossAxisAlignment.start,
98                      children: [
99                        Row(children: [
100 >                       Text(goal.categoryEmoji, ···
102                         const SizedBox(width: 8),
103 >                       Text( ···
107                         ),
108                       ]),
109                       const SizedBox(height: 4),
110 >                     Hero( ···
123                       ),
124                       const SizedBox(height: 8),
125                       Row(children: [
126                         const Icon(Icons.calendar_today,
127                             color: Colors.white70, size: 14),
128                         const SizedBox(width: 4),
129 >                       Text( ···
133                         ),
134                         const SizedBox(width: 16),
135 >                       Text( ···
139                         ),
140                       ]),
141                       const SizedBox(height: 8),
142                       ClipRRect(
143                         borderRadius: BorderRadius.circular(4),
144                         child: LinearProgressIndicator(
145                           value: goal.progressPercent,
146                           backgroundColor: Colors.white24,
147                           valueColor: const AlwaysStoppedAnimation(Colors
                              white),
148                           minHeight: 4,
149                         ),
150                       ),
151                     ],
152                   ),
153                 ),
154               ),
155             ),
156           ),
157           leading: IconButton(
158             icon: const Icon(Icons.arrow_back_ios, color: Colors.white),
159             onPressed: () => Navigator.pop(context),
160           ),
161           backgroundColor: theme.gradientHeader[0],
162         ),
```



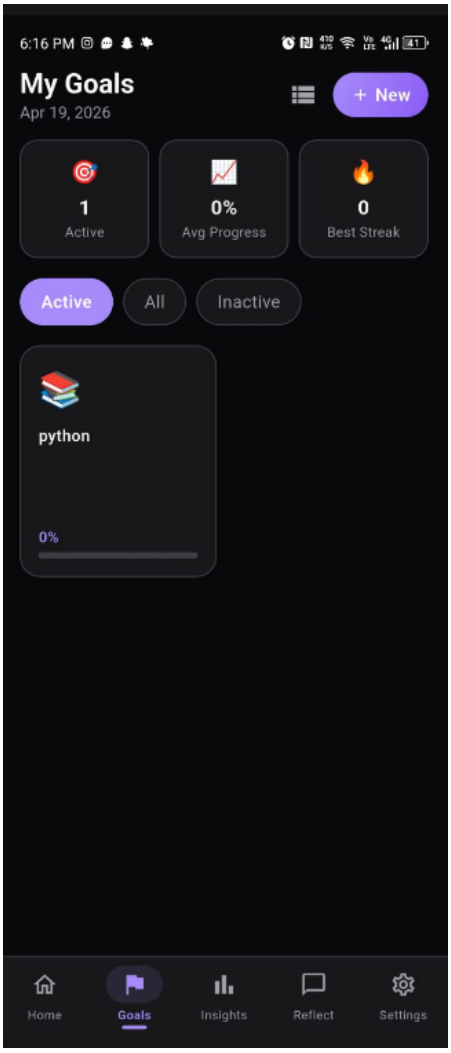
Goal screen

```
// Stats
SliverToBoxAdapter(
  child: Padding(
    padding: const EdgeInsets.fromLTRB(16, 8, 16, 16),
    child: Row(
      children: [
        _StatCard(theme: theme, label: 'Active', value: '$activeCount', icon: '🎯'),
        const SizedBox(width: 8),
        _StatCard(theme: theme, label: 'Avg Progress', value: '${avgProgress.toStringAsFixed(0)}%', icon: '📈'),
        const SizedBox(width: 8),
        _StatCard(theme: theme, label: 'Best Streak', value: '$bestStreak', icon: '🔥'),
      ],
    ),
  ),
),
```



Goal screen

```
49    return Scaffold(
50      backgroundColor: theme.background,
51      body: CustomScrollView(
52        slivers: [
53          //Floating app bar
54          SliverAppBar(  ···
104         ),
105
106         // Stats Row
107         SliverToBoxAdapter( ···
120         ),
121
122         // Filter pills
123         SliverToBoxAdapter( ···
154         ),
155         const SliverToBoxAdapter(child: SizedBox(height: 16)),
156         if (filtered.isEmpty)
157           // Empty state
158           SliverToBoxAdapter( ···
183           )
184         else if (_isGrid)
185           // GRIDVIEW — 2-column grid layout
186           SliverPadding( ···
250           )
251         else
252           // list view (AnimatedSwitcher)
253           SliverToBoxAdapter( ···
279           ),
280         // bottom spacing
281         const SliverToBoxAdapter(child: SizedBox(height: 32)),
282       ],
283     ),
284   );
```



Silvers:

- Goal detail: header is 160px tall expanded
- collapses to small bar as you scroll down, gradient always visible.
- Entire Goals screen scrolls as one app bar, stats, filters, and goal list all part of one scroll physics.